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Proficiency versus lexical processing efficiency as a measure of L2 lexical quality: Individual differences in word-frequency effects in L2 visual word recognition

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Abstract

This study investigated Korean-English second language (L2) speakers' recognition of high- and low-frequency English words and compared two individual difference measures in their role of representing lexical quality in L2: cloze test scores and inverse efficiency scores (IES; response latency corrected for the amount of errors committed), obtained from lexical decision on a separate set of words. Cloze test scores aimed to assess general L2 proficiency, whereas IES was purported to measure lexical processing efficiency. 109 adult Korean-English L2 speakers participated in the study. Results showed significant main effects of word frequency, cloze test scores, and IES on lexical decision times, replicating previous findings and confirming the predictions of the lexical quality hypothesis. Crucially, IES was revealed to be a better measure of individual differences in L2 lexical quality than were cloze test scores. These findings suggest that lexical quality (which can be operationalized in terms of online lexical processing efficiency) comprises a distinct subdomain of language skills on its own, which cannot be measured in full using conventional language proficiency tests.

Keywords L2 word recognition · Lexical quality · Word frequency · Cloze test · Inverse efficiency score

Introduction

Word frequency is one of the most widely studied linguistic factors that affect lexical processing. Words that occur with high frequency in a given language (e.g., *life* in English) are recognized more easily and rapidly than words with lower frequency (e.g., *dove*) (Brysbaert et al., 2011, 2018; Howes & Solomon, 1951; Monsell et al., 1989). According to the lexical quality hypothesis (Perfetti & Hart, 2002; Perfetti, 2007), the word-frequency effect arises from the fact that words can vary in the quality of their memory representations. A high-quality lexical representation has fully specified orthographic and phonological representations, which are robustly and flexibly linked to its meaning representation. On the contrary, a word representation has low lexical quality if its form and meaning components are not precise or flexible. Lexical quality has consequences for lexical processing such that high-quality lexical memory representations require less cognitive energy and time to be identified compared to low-quality representations. High frequency of a word entails that language users are repeatedly exposed to this word a number of times, which improves the quality of its representation in the lexicon. In turn, the word is recognized more quickly and with less effort compared to low-frequency words with low-quality representations.

A crucial assumption of the lexical quality hypothesis is that lexical quality is specific to the reader and to the word at the same time. In other words, lexical quality varies not only among words in a given reader's mental lexicon but also among individual readers for a given lexical item. One reader may have a high-quality representation for a given word and thus process it easily, while another reader may have a lower-quality representation for the same lexical item and require a relatively longer time to process it. Consequently, different

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readers may exhibit different frequency effects even to the same set of words. Indeed, individual differences in word-frequency effect are well documented in the literature (Ashby et al., 2005; Chateau & Jared, 2000; Kuperman & Van Dyke, 2011; Tainturier et al., 1992). For example, Chateau and Jared (2000) compared the effects of word frequency on lexical decision by participants with high versus low exposure to print. The results indicated a larger frequency effect on response times for the low print-exposure group than for the high print-exposure group. Similarly, Tainturier et al. (1992) compared two groups of native French speakers differing in education level (18 years vs. 11 years on average) in their French lexical decision times. They also found that the magnitude of frequency effect was smaller for the participants with longer years of education than for those with shorter years of education. In the same vein, Kuperman and Van Dyke (2011) measured participants' verbal skills with a variety of tests including word-naming and word-identification tests and found that these test scores modulated the effect of word frequency on fixation times during sentence reading such that the size of the frequency effects was weaker for higher scorers than for lower scorers in the verbal skill tests. These studies point towards the same relation between individual differences in language skills and word frequency; less skilled readers tend to show a bigger word-frequency effect than more skilled readers.

Another factor that is known to modulate word-frequency effects is bilingualism and language dominance. Research targeting unbalanced bilinguals indicated a greater frequency effect in their non-dominant language (or second language; L2) than in their dominant language (or first language; L1) (Cop et al., 2015; Duyck et al., 2008; Gollan et al., 2008; Mor & Prior, 2020). In addition, many studies showed that L2 speakers tend to show a bigger frequency effect than monolingual speakers of the given language in word identification and production tasks (Diependaele et al., 2013; Duyck et al., 2008; Gollan et al., 2008, 2011; Lehtonen et al., 2012; Lemhöfer et al., 2008; Whitford & Titone, 2012). Similar to the case of individual differences in reading skills, this language difference is mainly driven by low-frequency words; while L2 lexical processing is disadvantaged in general, this disadvantage is more prominent for low-frequency words than for high-frequency words. In their lexical entrenchment account, Diependaele et al. (2013) explains the larger word-frequency effect in L2 by referring to the lexical quality hypothesis. Because L2 speakers generally have limited experience in the target language compared to native speakers, their L2 lexical representations (especially low frequency ones) are less precise, in the sense that the spelling and sound codes of the words are not as accurate and the integration of semantic and formal representations is weaker. According to this view, a greater word-frequency effect in L2 is a result of limited experience (i.e., reduced lexical entrenchment) in the language rather than a result of being bilingual per se.

In addition to the effect of language dominance, this account makes another critical prediction that increased L2 experience would improve the overall quality of L2 lexical representations, resulting in a reduced word-frequency effect. This prediction has also been confirmed by earlier findings. Whitford and Titone (2012) reported that an increase in bilingual speakers' self-reported L2 exposure led to a reduced word-frequency effect during natural paragraph reading. Mor and Prior (2020) examined the influence of L2 reading fluency test scores, L2 vocabulary knowledge test scores, and self-reported exposure on frequency effects in L2 lexical decision. Their results indicated that among these variables, L2 vocabulary knowledge was the only significant predictor of L2 frequency effects. Diependaele et al. (2013) used the Lexical Test for Advanced Learners of English (LexTALE) vocabulary test (Lemhöfer & Boersma, 2012) as a measure of overall L2 proficiency and found that participants with high LexTALE scores showed a reduced frequency effect, which was consistently observed among speakers with three different L1 backgrounds (Dutch, French, and German). In a slightly different vein, Cop et al. (2015) found a modulating effect of L1 LexTALE scores on L2 word-frequency effects: an increase in L1 LexTALE scores led to a smaller frequency effect in L2. In interpreting this result, the authors speculate that L1 LexTALE scores might have measured an abstract reading skill or general language aptitude. They also claim that this speculation concurs with the lexical quality hypothesis in that both views acknowledge individual differences among readers in the abilities to learn and process words (citing Perfetti et al., 2005). In sum, with previous studies adopting inconsistent individual difference measures in L2 (e.g., exposure to L2, L2 vocabulary knowledge, or overall L2 proficiency), it still remains inconclusive what exactly it is that modulates L2 word-frequency effects.

Finding an answer to this question is of particular importance in L2 studies, because one's general proficiency in L2 (a concept that encompasses not only vocabulary knowledge but also phonological, morphosyntactic, and semantic knowledge as well as communicative abilities) and lexical quality in L2 can be independent of each other depending on speakers' learning environments and strategies. Thus, under the hypothesis that L2 lexical quality modulates word-frequency effects in L2 lexical processing, the current study aimed to identify a proper measure of L2 lexical quality by comparing the effects of two L2-related factors. The first factor is general L2 proficiency measured using an English cloze test. Cloze tests are widely used in second-language studies as a measure of general proficiency and are known to correlate with other standardized proficiency test scores (e.g., Chae & Shin, 2015; Ionin et al., 2013; Lee, 1997; Tremblay & Garrison, 2008; Tremblay, 2011). The second factor is online word-processing efficiency measured by a timed lexical decision task. Lexical decision tasks have been used widely for decades to assess word recognition and lexical access in both monolingual (e.g., Bauer & Stanovich, 1980; Perea & Pollatsek, 1998) and bilingual (Soares & Grosjean, 1984; Gerard & Scarborough, 1989) contexts. Earlier studies on individual differences in language processing have shown time-sensitive measures such as timed lexical decision or rapid naming correlate with other language skills (e.g., Araújo et al., 2015; Katz et al., 2012; Kuperman & Van Dyke, 2011; Torgesen et al., 1997). Furthermore, not only response accuracy but also response speed is argued to represent the efficiency of lexical processing, particularly in L2 (Harrington, 2006; Segalowitz & Segalowitz, 1993; Segalowitz et al., 1998; Van Hell & Dijkstra, 2002). We thus adopted these two measuring tools to compare the role of general L2 proficiency and online L2 word-processing efficiency in modulating word-frequency effects in L2 visual word recognition.

This comparison will shed light on the nature of lexical quality in L2. If we hypothesize that L2 lexical quality is a concept independent from general proficiency in L2, it is predicted that readers' online L2 word-processing efficiency, as measured by a timed lexical decision task, would be a better mediator of L2 word-frequency effects than cloze test scores. This line of results will suggest that lexical quality in L2 comprises a distinct subdomain of language skills on its own, which cannot be measured in full using conventional language proficiency tests. If, on the other hand, the result shows that cloze test scores play a better role in mediating frequency effects than timed lexical decision task scores, or if the two measures do not differ from each other in their effects, these findings would indicate that the concept of lexical quality is essentially inextricable from general language proficiency at least in the non-dominant language of L2 speakers.

In addition, the current study targeted L2 speakers whose L1 and L2 adopt different writing systems. Most previous studies on L2 frequency effects examined bilingual speakers of two languages with similar scripts, such as Dutch, French, Spanish, German, and English. When bilingual speakers of similar-script languages perform a visual word identification task in their L2, their L1 lexicon may also come into play in the identification process, and this language-nonselective lexical access may yield an additional influence on the frequency effect (see Gollan et al., 2008, for detailed discussion on this issue). The only study that investigated individual differences in L2 frequency effects in different-script L2/bilingual speakers, to our knowledge, was conducted by Mor and Prior (2020), who found a significant interaction between word frequency and L2 vocabulary knowledge in Hebrew-English bilingual speakers' English lexical decision. Their finding is evidence to the claim that the variability in the frequency effect in L2 is *not* a result of cross-language lexical activation/competition, but rather a result of varying levels of vocabulary knowledge in L2. We aimed to add to this scarce literature by examining individual differences in Korean-English L2 readers' visual recognition of high- and low-frequency English words.

Method

Materials

Lexical decision task materials

A list of 30 high-frequency and 30 low-frequency English words was selected based on the English Lexicon Project (Balota et al., 2007). The SUBTLEX frequency for the high-frequency words ranged from 509 (*baby*) to 4,008 (*right*) per million, with an average of 929.1 ($SD = 714.9$). The low-frequency words had SUBTLEX frequency from 4 (*crow*) to 10 (*skirt*), with an average of 6.3 ($SD = 1.3$). The average number of letters per word was 4.7 ($SD = 0.4$) for both high- and low-frequency groups, and all words were either mono- or di-syllabic. We treated word frequency as a binary factor rather than a continuous one in order to obtain a clearer difference between high- and low-frequency words.

Since the corpus-based frequency of words may not necessarily apply to L2 learning contexts, we conducted a familiarity rating survey on the words to ensure that their frequency also represented their familiarity to L2 speakers. Fifty-three Korean learners of English, none of whom participated in the main task, were invited to rate how familiar they felt with each word on a 7-point scale (1: *not at all familiar* – 7: *very familiar*). Results of the survey indicated that the words with high SUBTLEX frequency received familiarity ratings of 6.7 on average ($SD = 0.2$), and those with low frequency received ratings of 5.0 on average ($SD = 1.2$). This difference was statistically significant ($t = 7.7$, $p < .001$), confirming that the frequency difference between the two sets of words was applicable to Korean-English L2 speakers' English lexicon. The semantic concreteness and age of acquisition (AOA) of the target items were obtained from the English Lexicon Project. The average concreteness was 3.64 ($SD = 1.19$) and the average AOA was 5.72 ($SD = 1.87$). The two frequency groups did not significantly differ in concreteness ($t = 0.33$, $p = 0.74$) or AOA ($t = 1.06$, $p = 0.30$). A full list of target items and their frequency is given in the Appendix.

In addition to the target stimuli, 120 English words were included in the lexical decision task to be used to measure online lexical processing efficiency (see the *Lexical processing efficiency measure* section below). The task also included 180 English nonwords extracted from the English Lexicon Project. In sum, the task consisted of lexical decision trials on 180 English words and 180 nonwords.

Lexical processing efficiency measure

We operationalized participants' lexical processing efficiency in L2 in terms of their performance in timed lexical decision on English words. In order to do so, 120 English words, none of which appeared in the target list, were added to the stimulus set in the lexical decision task. The motivation behind using one word set for the lexical decision task and another to assess online lexical processing efficiency was to avoid a situation in which the dependent variable (lexical decision times) and one of the predictors (online lexical processing efficiency) come from the same data points. The 120 words varied in the number of letters (M = 5.2, SD = 1.1) and SUBTLEX frequency per million (M = 235.2, SD = 519.3). Each participant's mean response time (RT) in correct trials in milliseconds and the proportion of errors (PE) were used to obtain the inverse efficiency score (IES; Townsend & Ashby, 1978). The IES is defined as $IES = \frac{RT}{1-PE}$ (i.e., a participant's response latency corrected for the accuracy rate), and it is the oldest and most frequently used measure that integrates RT and accuracy (Vandierendonck, 2017). More rapid responses and a smaller number of errors in the lexical decision task will result in a lower IES, indicating more efficient lexical processing.

English proficiency measure: A cloze test

As a measure of participants' general English proficiency, we used a multiple-choice cloze test adopted from Chae and Shin (2015) and Ionin et al. (2013), which was based on a passage taken from *American Kernel Lessons: Advanced Students' Book* (O'Neil et al., 1981). In the passage, every seventh word was removed and replaced by three choices including only one choice appropriate for the context. The number of items to which a participant provided the correct answer was used as their English proficiency score (maximum score = 40). Ionin et al. (2013) reported a Cronbach's alpha of .817 for the test in their study involving Spanish learners of L2 English, showing a high internal consistency. Once data collection was complete, participants' cloze test scores were converted to z-scores using their grand mean and standard deviation.

Participants

One hundred and nine adult Korean speakers living in Korea were recruited to participate in the task through the Internet. Twelve participants who had lived abroad for more than a year or who did not identify themselves as either native Korean speakers or English L2 speakers were excluded from the analysis. One participant was excluded due to noticeably long RTs (over 3 s for more than 20% of the trials). The data from the remaining 96 participants, defined as Korean speakers of English as L2, were included in the analysis (63 females, 33 males). They were 23 years old on average at the time of participation (SD = 4.1), and their average age when they had first started learning English was 8.1 years (SD = 2.4).

Procedure

A lexical decision task was built using PsychoPy3 (Peirce et al., 2019) and uploaded to the Pavlovia.org website to enable online data collection. Once the participant accessed the experiment page on their own computer, they were first asked to provide informed consent. They then proceeded to the lexical decision task, during which they decided the lexicality of each stimulus by pressing the 'a/A' key for words or the 'l/L' key for nonwords on their keyboard. Each trial began with a fixation cross (+) appearing at the center of the screen, which was replaced with a stimulus item after 1 s. The stimulus stayed on the screen until the participant made a response. The task started with ten practice trials, during which the participant received feedback on the correctness of their responses, and afterwards, the 180 words and 180 nonwords were presented in a randomized order. The stimuli were presented in white san serif font (Nanum Gothic) on a black background. The height of the letters was set to 10% of the height of the screen. After the lexical decision task, the participants proceeded to the cloze test. The cloze test was conducted using Google Form. The entire session took approximately 30 min.

There was no time limit to finish the cloze test. While both timed and untimed cloze tests have been used in the literature (Chae & Shin, 2015; Ionin et al., 2013; Tremblay, 2011), we used an untimed format for two reasons. First, it was our aim to assess individual participants' overall language competence, rather than time-constrained language performance. Second, since imposing a time limit may introduce confounding factors such as test-taking strategies and test anxiety (Amiryousefi & Tavakoli, 2011; Onwuegbuzie & Seaman, 1995; Opperman, 2020), an untimed cloze test score was expected to be a more valid measure of language competence itself.

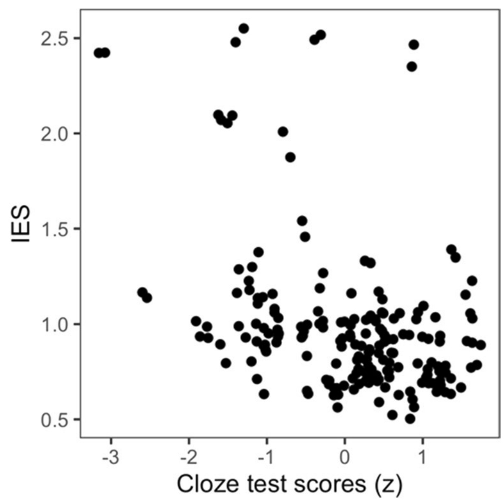

Fig. 1 Individual participants' cloze test scores and inverse efficiency scores (IES)

Statistical analysis

Accuracy and RT of each response were measured during the task. Response accuracy was coded as a binary measure (accurate = 1, inaccurate = 0), and RT was measured as the time in milliseconds between the onset of stimulus presentation and a response. Trials on which RT was longer than 3,000 ms were excluded from the analysis (118 trials). The cut-off for trial filtering was set to 3,000 ms following Mor and Prior (2020), which is higher than those used in typical lexical decision studies, because the test language was the second language of our participants, as was the case in Mor and Prior (2020). Effects of word frequency and individual difference measures on lexical decision performance were analyzed using logistic mixed-effects regression models for response accuracy and linear mixed-effects regression models for RT. The models were built using the glmer()/lmer() functions from the *lme4* package (Bates et al., 2015) in R, version 4.0.3 (R Core Team, 2023).

Three sets of mixed-effects regression models were built by including different individual difference variables as fixed effects. The first models (each on accuracy and RT) included word frequency, cloze test scores, and the interaction between these two as fixed effects. Results of these models are reported in the *Results of analysis with cloze test scores* section. The second models (each on accuracy and RT) included word frequency, IES, and their interaction as fixed effects. Results of these models with IES are also reported in the *Results of analysis with IES* section. Finally, a full model was built on RT data by including both the IES and the cloze test scores and their interactions with frequency, in addition to a main effect of word frequency, as fixed effects. Since semantic features of the words may also affect lexical decision times, all models additionally included concreteness and AOA of words as controlling factors. Explanatory powers of the first two models on RT were then compared to that of the full model using the anova() function in R. The amount of variance explained by each model (R^2) was obtained using the r2() function of the *performance* package. Results of these model comparisons are reported in the *Model comparisons* section. All models included individual difference measures as continuous variables and word frequency as a contrast-coded binary factor (High = 1, Low = -1). For random effects, we started with the maximal structure with by-participant and by-item random intercepts as well as random slopes for frequency over participants and individual differences over items (Barr et al., 2013), but not all models reached convergence with this structure. In order to make the models comparable only with regard to fixed effects of individual difference variables, we selected the maximal random structure that allowed all models to converge, which consisted of by-participant and by-item random intercepts.

Results

Cloze test scores and inverse efficiency scores (IES)

Since we obtained two individual difference measures for participants, cloze test scores and IES, we first examined how correlated these two measures were. Individual participants' cloze test scores and IES are illustrated in Fig. 1.

The Pearson correlation coefficient between the two measures was -0.406 ($p < 0.001$), indicating that those who

Table 1 Descriptive statistics of lexical decision accuracy (error rates) and response time (RT) by participants' cloze test scores (median-split at 30.5) and target word frequency

Group	Cloze test score		Error rates				RT (ms)				
			Low frequency		High frequency		Low frequency		High frequency		Frequency effect
	Mean	*SD*	Mean	*SD*	Mean	*SD*	Mean	*SD*	Mean	*SD*	
Low cloze ($N = 48$)	25.8	3.6	0.25	0.43	0.04	0.19	920	435	739	329	181
High cloze ($N = 48$)	33.5	2.1	0.19	0.39	0.03	0.17	865	391	679	287	186

Table 2 Estimated effects of participants' cloze test scores and word frequency on lexical decision accuracy and response time (RT)

	Accuracy				RT			
	Model: Accuracy ~ (Frequency * Cloze) + Concreteness + AOA + (1|Participant) + (1|Item)				Model: RT ~ (Frequency * Cloze) + Concreteness + AOA + (1|Participant) + (1|Item)			
	β	*SE*	*z*	*Pr(>|z|)*	*β*	*SE*	*t*	*Pr(>|t|)*
(Intercept)	4.963	0.933	5.320	0.000 ***	601.448	79.225	7.592	0.000 ***
Frequency	0.691	0.216	3.195	0.001 **	-57.992	18.025	-3.217	0.002 **
Cloze	0.174	0.097	1.794	0.073 .	-45.350	16.658	-2.722	0.008 **
Concreteness	-0.173	0.148	-1.167	0.243	13.964	11.989	1.165	0.249
AOA	-0.281	0.091	-3.088	0.002 **	27.700	8.016	3.456	0.001 **
Frequency:Cloze	-0.092	0.058	-1.586	0.113	3.528	4.654	0.758	0.448

*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$

were more proficient in English tended to be more efficient in processing English words. However, the correlation coefficient was only of a moderate size, which suggests that these two variables represented two abilities that are different from each other, although overlapping to some degree: general English proficiency and online lexical processing efficiency, respectively. It thus seemed valid to compare them in terms of their roles in modulating a word-frequency effect in lexical decision, as an attempt to identify a more appropriate measure of L2 lexical quality. The following sections report on the effects of each measure on lexical decision accuracy and latency of low- and high-frequency word lexical decision.

Results of analysis with cloze test scores

Table 1 shows descriptive statistics of lexical decision accuracy and RT for high- and low-frequency words by participants' cloze test scores. Participants are median-split into two cloze test score groups in Table 1 to summarize the results, although cloze test scores were treated as a continuous variable in the analysis. Overall, participants with higher cloze test scores tended to be more accurate and faster in recognizing the target words regardless of word frequency. Also, high-frequency words were recognized more accurately and rapidly than low-frequency words.

The results of mixed-effects regression analyses on the effects of word frequency (Low vs. High) and participants' cloze test scores (treated as a continuous variable) on accuracy and RT are summarized in Table 2.

The main effect of word frequency was significant in both accuracy and RT analyses. High-frequency words were recognized more accurately and rapidly than were low-frequency words, which demonstrates a typical word-frequency effect on lexical decision. The main effect of cloze test scores on RT was also significant; participants with higher cloze test scores responded faster than those with lower cloze test scores. Cloze test scores did not have a significant influence on lexical decision accuracy, as all participants exhibited fairly high accuracy rates. Crucially, the interaction between frequency and cloze test scores was not significant in either accuracy or RT analyses. This result runs against previous findings and the prediction of the lexical quality hypothesis that less proficient L2 speakers tend to exhibit a greater word-frequency effect than do more proficient speakers.

Results of analysis with IES

Descriptive statistics of lexical decision accuracy and RT for high- and low-frequency words by participants' IES is

Table 3 Descriptive statistics of lexical decision accuracy (error rates) and response time (RT) by participants' inverse efficiency scores (IES) (median-split at 0.90) and target word frequency

Group	IES		Error rates				RT (ms)				
			Low frequency		High frequency		Low frequency		High frequency		Frequency effect
	Mean	*SD*	Mean	*SD*	Mean	*SD*	Mean	*SD*	Mean	*SD*	
High IES ($N = 48$)	1.24	0.47	0.23	0.42	0.03	0.17	1028	477	794	368	234
Low IES ($N = 48$)	0.75	0.09	0.22	0.42	0.04	0.19	760	286	624	207	136

Table 4 Estimated effects of participants' inverse efficiency scores (IES) and word frequency on lexical decision accuracy and response time (RT)

	Accuracy				RT			
	Model: Accuracy ~ (Frequency * IES) + Concreteness + AOA + (1|Participant) + (1|Item)				Model: RT ~ (Frequency * IES) + Concreteness + AOA + (1|Participant) + (1|Item)			
	β	*SE*	*z*	*Pr(>|z|)*	*β*	*SE*	*t*	*Pr(>|t|)*
(Intercept)	4.739	0.968	4.897	0.000 ***	322.872	84.142	3.837	0.000 ***
Frequency	0.395	0.277	1.425	0.154	-24.625	21.134	-1.165	0.247
IES	0.247	0.269	0.918	0.359	276.255	30.239	9.136	0.000 ***
Concreteness	-0.172	0.148	-1.167	0.243	14.107	11.993	1.176	0.245
AOA	-0.281	0.091	-3.086	0.002 **	27.934	8.018	3.484	0.001 ***
Frequency:IES	0.318	0.184	1.730	0.084 .	-33.058	11.048	-2.992	0.003 **

*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$

shown in Table 3. Participants are median-split into two IES groups in Table 3 to summarize the results, although IES was treated as a continuous variable in the analysis. Participants with lower IES (representing more efficient word processing) tended to be faster than those with higher IES, although there was no noticeable difference in accuracy. Also, high-frequency words were recognized more accurately and rapidly than low-frequency words.

The results of mixed-effects regression analyses on the effects of word frequency (Low vs. High) and participants' IES (treated as a continuous variable) on accuracy and RT are summarized in Table 4.

As in the analyses with cloze test scores, the main effect of word frequency was significant in both models. Responses to high-frequency words were more accurate and faster than responses to low-frequency words, replicating a word-frequency effect reported in prior research. The main effect of IES on RT was also significant, as participants with lower IES (representing higher efficiency) made more rapid responses than those with higher IES. The effect of IES was not significant in the accuracy analysis, presumably due to fairly high accuracy rates overall. Importantly, IES was found to significantly interact with word frequency to affect RT; the effect of word frequency on RT was greater among less efficient participants than among more efficient participants. This interaction effect contrasts with the lack of a significant interaction between word frequency and cloze test scores (Table 2), and directly confirms the prediction of the lexical quality hypothesis that readers with lower lexical quality would show a greater word-frequency effect than those who possess higher lexical quality.

Model comparisons

The results reported in the preceding sections indicated that only IES, not cloze test, scores significantly interacted with word frequency to affect lexical decision times. Figure 2 visualizes the interaction effects between cloze test scores and word frequency and between IES and word frequency estimated by the mixed-effects models.

We compared the role of cloze test scores and IES in mediating a word-frequency effect in Korean speakers'

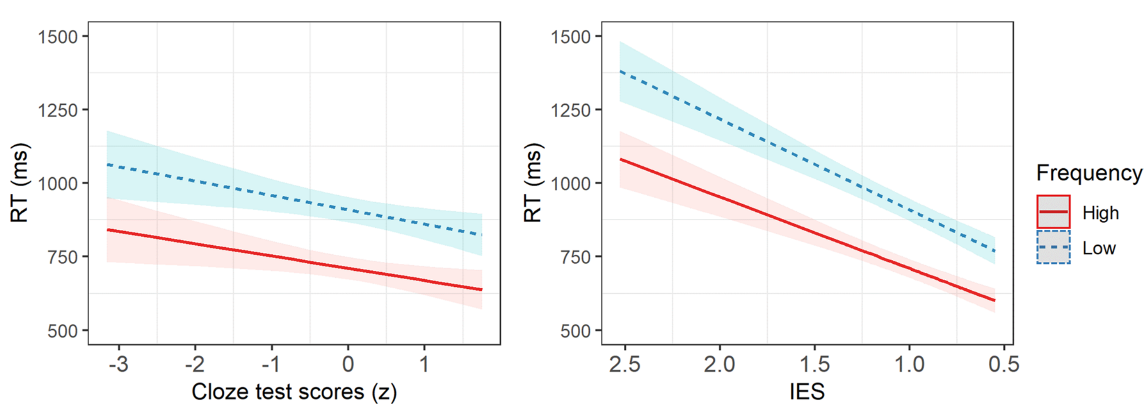

Fig. 2 Lexical decision response time (RT) as a function of word frequency and participants' cloze test scores (**left**) and inverse efficiency scores (IES) (**right**) as predicted by mixed-effects models. IES is an inverse measure of lexical processing efficiency, with smaller numbers representing higher efficiency

Table 5 Model comparisons. Comparison 1 compares the full model with a nested model with only cloze test scores, and Comparison 2 compares the full model with a nested model with only inverse efficiency scores (IES)

Model comparison 1									
Full model: RT ~ (Frequency * (IES + Cloze)) + Concreteness + AOA + (1|Participant) + (1|Item)									
Nested model: RT ~ (Frequency * Cloze) + Concreteness + AOA + (1|Participant) + (1|Item)									
	npar	R^2	AIC	BIC	logLik	deviance	χ^2	Df	Pr(>χ^2)
Nested model	9	0.29	71019	71078	-35501	71001			
Full model	11	0.29	70962	71033	-35470	70940	61.179	2	0.000 ***
Model comparison 2									
Full model: RT ~ (Frequency * (IES + Cloze)) + Concreteness + AOA + (1|Participant) + (1|Item)									
Nested model: RT ~ (Frequency * IES) + Concreteness + AOA + (1|Participant) + (1|Item)									
	npar	R^2	AIC	BIC	logLik	deviance	χ^2	Df	Pr(>χ^2)
Nested model	9	0.29	70958	71017	-35470	70940			
Full model	11	0.29	70962	71033	-35470	70940	0.4485	2	0.7991

English lexical decision through model comparisons. First, a full mixed-effects model on RT was built by including both cloze test scores and IES as individual difference variables. The explanatory power of this full model was then compared with those of the two nested models: one with only cloze test scores (reported in Table 2) and the other with only IES (reported in Table 4). The results of the two comparisons are summarized in Table 5.

The results of the first comparison show lower AIC and BIC values for the full model than for the nested model, which indicates that the full model explains more variability in the data than the nested model. The statistical significance of χ^2 suggests that including IES as an additional fixed effect (as in the full model) substantially increased the explanatory power of the model compared to having only cloze test scores (as in the nested model). On the contrary, in the second comparison, the lower AIC and BIC values for the nested model than for the full model are indications of the fact that the nested model explains more variability of the data than does the full model. The lack of statistical significance of χ^2 indicates that adding cloze test scores to the model as another fixed effect (as in the full model) did not significantly improve the model fit compared to having IES as the only individual difference variable (as in the nested model). Since the two models do not differ in their explanatory power, the relative parsimony of the nested model makes it favorable over the full model. These results together stand in favor of IES over cloze test scores as an individual difference variable mediating the effect of word frequency on lexical decision.

Discussion

This study examined Korean-English L2 readers' recognition of high- and low-frequency English words and compared two individual difference measures, each assessing general L2 proficiency and L2 word-processing efficiency, respectively, in their role of representing individual differences in L2 lexical quality. First of all, the results indicated a significant main effect of word frequency on lexical decision accuracy and RT. High-frequency target words were recognized more accurately and rapidly than low frequency target words. According to the lexical quality hypothesis (Perfetti & Hart, 2002; Perfetti, 2007), this word-frequency effect arises from variabilities in the quality of lexical representations. In the reader's lexicon, words that occur frequently in the language have more specific and redundant orthographic and phonological representations compared to less frequently occurring words. Consequently, the recognition of more frequent words can take place more automatically and efficiently. To put it in terms of a similar usage-based account of lexical entrenchment, more frequent use of a word results in more precise and strongly integrated lexical memory representations for the word, which in turn reduces the amount of energy required to process it (Diependaele et al., 2013).

Also, both individual difference variables – cloze test scores and IES – were found to have a significant effect on lexical decision latency. Higher cloze test scores and lower IES were associated with faster RTs. This suggests that, although the cloze test and the IES were aimed to assess general proficiency and lexical processing efficiency respectively, the two variables were both measuring skills that have an impact on the word recognition process. In addition to the main effects, IES also significantly interacted with word frequency to affect RTs. This interaction has likely been driven by less efficient readers (with high IES) being slowed down in responding to low-frequency words to a greater extent than were more efficient readers. This result conforms to the lexical quality account, which predicts less experienced or less skilled readers to exhibit a large word-frequency effect because they enjoy a full benefit of additional exposure to frequently occurring words. In contrast, more experienced or skilled readers have word representations whose lexical quality is already close to the ceiling level, especially for high-frequency words, and as a result, additional exposure to these frequently occurring words compared to low-frequency words does not make such a prominent difference for these readers. In other words, skilled readers benefit less from any additional encounter with a word, resulting in a smaller RT difference between high- and low-frequency words.

In contrast to IES, cloze test scores failed to yield such an interaction effect with word frequency. Furthermore, having IES as the only individual difference variable led to the model that explains larger variability of the data than the model with only cloze test scores or that including both individual difference variables. These two variables compared in this study differ in two respects: the timedness of the task and the breadth of the competence that they aim to assess. First, the cloze test was an untimed task, so its scores did not reflect time-sensitive performance efficiency, whereas IES was a time-sensitive measure by definition. The result favoring IES over cloze test scores thus suggests that lexical quality is better represented by a time-sensitive measure. This corroborates the view of the lexical quality hypothesis that a high-quality representation results in more efficient (i.e., easier and faster) word retrieval than a low-quality representation (Perfetti & Hart, 2002). Secondly, the cloze test was purported to assess learners' general language proficiency in a broad sense (including lexical and morphosyntactic knowledge as well as discourse competence), as evidenced by its high correlation with scores from standardized tests such as TOEFL (Tremblay & Garrison, 2008; Tremblay, 2011). IES in this study, in contrast, was obtained based on the participants' performance in lexical decision, and thus it was more specific to abilities to recognize visual words presented in isolation. Hence, our results indicate that lexical quality comprises a factor that is distinguished from general language proficiency and more specific to word knowledge. This interpretation is consistent with Mor and Prior's (2020) finding that L2 vocabulary knowledge, but not L2 reading fluency or self-reported exposure, mediated frequency effects in the L2 lexical decision.

What still makes it tricky to disentangle lexical quality from general proficiency is previous research reporting a significant interaction between general proficiency and frequency in L2 word recognition (Cop et al., 2015; Diependaele et al., 2013; Whitford & Titone, 2012). For example, in Diependaele et al.'s (2013) analysis on L2 word-identification latency, participants' LexTALE scores were found to interact with word frequency. Whitford and Titone (2012) also reported that participants' self-reported L2 exposure mediated the effect of word frequency in paragraph reading. A critical difference between these prior studies and the present study, in our view, is that these earlier studies targeted bilinguals of similar-script languages (e.g., Dutch-English, French-English, German-English, English-French), whereas our participants' L1 (Korean) and L2 (English) adopt different writing systems. Due to cross-linguistic orthographic similarity, L2 word recognition in the earlier studies might have been influenced not only by L2 lexical quality but also by external factors such as language-nonselective activation of orthographically related L1 words (Gollan et al., 2008) or general language aptitude closely tied to their L1 experience (Cop et al., 2015). In contrast, the L2 speakers in the current study began to learn and use Roman scripts only as part of their English learning. Thus, their English word recognition must have hardly been affected by language-nonselective lexical access or L1 experience, at least to a lesser degree than bilinguals of similar-script languages.

Furthermore, the lexical quality hypothesis posits that lexical quality is defined in part by the specificity of orthographic representations of words (Perfetti, 2007). It follows that whether two languages of a bilingual speaker use similar scripts or not must have a consequence on their L2 lexical quality. When L2 learners of similar-script languages learn L2 words, orthographic forms of the words are similar to their L1 writing system, making it relatively easier for them to develop more accurate and specific (i.e., high-quality) orthographic lexical representations. On the other hand, L2 learners of different-script languages need to become familiarized with an entirely new writing system to learn L2 words. As a result, the orthographic codes of L2 words in their lexicon are likely to be more slowly and less precisely established in general. In this sense, it can be argued that the current study tapped into the lexicon of L2 speakers who are relatively more novice in their orthographic knowledge than the bilingual speakers targeted in previous studies.

Although we interpret our findings as favoring IES over cloze test scores as a measure of L2 lexical quality, we do not intend to downplay the crucial role of language proficiency in L2 word recognition. Response latency in the lexical decision task was clearly dependent on the participants' cloze test scores, with more proficient speakers responding significantly faster than less proficient speakers. Their cloze test scores were also moderately correlated with IES, which indicates that the two variables are not entirely independent from each other. Rather, given the literature, it is our perspective that lexical quality (which can be operationalized as online lexical processing efficiency) comprises a subdomain of language skills on its own, although it is intricately linked to general language proficiency and its other subdomains. The inextricability of lexical quality from other language skills is also noted in Perfetti and Hart's (2002) proposal of the lexical quality hypothesis. Future research is necessary to reach a more precise and quantifiable definition of lexical quality and to explore diverse proper measuring tools.

It is also not our intention to suggest that IES stands as the only optimal measure of lexical processing performance in L2. Although we used IES to represent the efficiency of lexical processing, it is by no means the only performance score that integrates speed and accuracy. Earlier studies have reported similar validities of other scoring methods, such as the rate residual score, the bin score, and the linear integrated speed-accuracy score (Hughes et al., 2014; Vandierendonck, 2017). However, these scoring methods refer to the difference in performance across more than one experimental conditions (e.g., in the task-switching paradigm), and therefore are only appropriate when one's interest lies in figuring out conditional differences. Since we did not have manipulated conditions of the stimuli and needed to obtain one efficiency value for each individual participant, IES was, to our knowledge, the only scoring method that was appropriate for the current study. Admittedly, IES itself is not free of limitations (e.g., see Bruyer & Brysbaert, 2011), and the result that IES is a significant predictor of RT is not completely unexpected since IES is a function of lexical decision times. Nevertheless, a crucial implication of our findings is that lexical quality in L2 can be best quantified using a scoring method that integrates speed and accuracy of lexical processing, as the quality of lexical representations has consequences in both. It is perhaps for the same reason that earlier studies on L1 lexical quality used tasks that are timed but also accuracy-oriented such as the Test of Word Reading Efficiency and the Nelson-Denny Test (e.g., Andrews et al., 2020; Luke et al., 2015; Taylor & Perfetti, 2016).

The present study collected lexical decision data and cloze test scores using a web-based experiment procedure. Recruited participants were provided with a URL to the experiment page, and they completed the tasks at their convenience using their own computer with no face-to-face interaction with an experimenter. Although this differs from traditional laboratory-based studies, in which participants perform tasks in a laboratory setting in the presence of an experimenter, the idea of web-based behavioral experiments is not new. A number of recent studies have documented efficiency and reliability of collecting behavioral data through web-based experiment or survey platforms (e.g., Aguasvivas et al., 2020; Anwyl-Irvine et al., 2020; Chetverikov & Upravitelev, 2016; Kim et al., 2021; Semmelmann & Weigelt, 2017). The fact that the current study successfully replicates preceding research findings on the effects of word frequency, L2 proficiency, and their interactions on L2 word recognition adds to this body of literature to substantiate the validity of the web-based experiment paradigm in psycholinguistic research.

Conclusion

To conclude, the current investigation sheds light on the nature of lexical quality as an individual difference variable that affects lexical processing in L2. We found that lexical quality can be properly operationalized in terms of how efficiently a given word input can be processed by a given reader. These findings provide evidence to support the use of a time-sensitive efficiency measure that is specific to word-recognition performance, rather than a general language proficiency test, as a tool for assessing individual differences in L2 lexical quality.

Appendix: Experiment materials

High-frequency words		Low-frequency words	
Word	Frequency*	Word	Frequency*
right	5.31	skirt	2.71
think	5.14	creek	2.66
never	4.84	laser	2.64
sorry	4.78	stove	2.59
thank	4.76	chili	2.58
love	4.76	armor	2.57
maybe	4.67	draft	2.57
help	4.67	jelly	2.56
these	4.66	shelf	2.55
night	4.65	hobby	2.55
first	4.63	tubes	2.53
great	4.62	peach	2.51
life	4.61	booty	2.50
still	4.60	cloth	2.49
those	4.58	debts	2.49
other	4.57	glue	2.48
stop	4.56	spine	2.47
name	4.52	clip	2.46
money	4.51	tomb	2.46
place	4.49	scope	2.46
kind	4.48	dove	2.46
hello	4.48	vest	2.46
years	4.46	lone	2.43
leave	4.46	buyer	2.42
girl	4.45	mode	2.42
three	4.44	waist	2.42
wrong	4.43	thorn	2.42
might	4.42	flick	2.41
house	4.42	hound	2.41
baby	4.42	crow	2.36
Mean (SD)	**4.61 (0.21)**	**Mean (SD)**	**2.50 (0.08)**

* Word frequency represents logged SUBTLEX frequency obtained from the English Lexicon Project (Balota et al., 2007)

Nonwords					
bics	pumb	bolks	drass	marbs	puses
birt	rown	brare	drine	masic	rudge
daid	runp	brune	frash	moost	scing
dras	sagi	burls	frunt	narns	slase
fias	suet	careb	gared	nello	snart

filo	vorm	caths	gevel	noast	sowel
foze	ahide	ceter	grive	pease	tards
glab	balgy	ciped	gutch	plock	thase
guse	blenk	croom	keach	powls	trink
luct	bocer	dayed	liets	poxer	voots

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Data availability Anonymized data, experiment materials, and analysis scripts for this manuscript are available at the Open Science Framework repository via https://osf.io/hrukc/?view_only=9e020f4a068247bca5db00e8def21d02.

Declarations

Ethics approval The methodologies used in this study were approved by the Institutional Review Board of Gwangju Institute of Science and Technology (approval number 20201008-HR-56-06-02).

Consent to participate and consent for publication Informed consent was obtained from all individual participants included in the study. This consent informed all individual participants regarding publishing their data.

Conflict of interest The authors declare that they have no potential conflicts of interest with respect to the research, authorship, or publication of this article.

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